

■ Discovering the Pythagoras Theorem!

A fun introduction for Grade 4 students

In this lesson you will learn about **right-angled triangles**, discover the three special sides, and unlock the famous formula **$a^2 + b^2 = c^2$** !

#	Topic	
1	Meet the Shapes	Right angles & triangles
2	The Three Sides	Hypotenuse, Side A, Side B
3	Squares on Each Side	Visual proof
4	The Theorem	$a^2 + b^2 = c^2$
5	Real-Life Example	The Ladder Problem
6	Step-by-Step Calculation	3-4-5 triangle
7	Practice Problems	Try it yourself!
8	Fun Quiz	Test your knowledge

■ Section 1: Meet the Shapes

Look around you — triangles, squares, and rectangles are everywhere! Your notebook corner, a sandwich cut diagonally, the ramp at the park — they all hide maths magic inside them.

■ Spot the triangle, square, and rectangle around your classroom right now!

What is a Right-Angled Triangle?

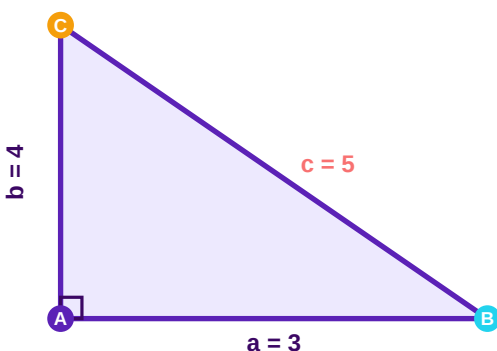
A **right-angled triangle** has one very special corner that looks exactly like the corner of your textbook. That corner is called a **right angle** and it measures exactly 90° . We mark it with a tiny square symbol (■) in the corner.

■ Tip: Look for the tiny square in the corner — that is your right angle marker!

■ Section 2: The Three Sides

Every right-angled triangle has three sides, and each side has its own special name and role. Learning their names is the first step to unlocking the Pythagoras Theorem!

Side Name	Description
Hypotenuse (c)	The longest side. Always sits opposite the right angle.
Side A (a)	One of the two shorter sides that form the right angle.
Side B (b)	The other shorter side that forms the right angle.



Here is a right-angled triangle with the sides labelled. The right angle is at corner **A** (bottom-left), the hypotenuse **c** goes from **B** to **C**.

■ Remember: The hypotenuse is **always the biggest side** in any right-angled triangle!

■ Section 3: Squares on Each Side

Here is something amazing! If you draw a square on *each* side of a right-angled triangle, the areas of the two smaller squares will always add up to the area of the biggest square.

Square on Side a	Square on Side b	Square on Hypotenuse c
Area = $a \times a = a^2$	Area = $b \times b = b^2$	Area = $c \times c = c^2$
e.g. $3 \times 3 = 9$	e.g. $4 \times 4 = 16$	e.g. $5 \times 5 = 25$

■ The two small squares (a^2 and b^2) together always equal the big square (c^2)!

■ Section 4: The Pythagoras Theorem

All of that amazing discovery leads to one beautiful, short formula that works for *every single* right-angled triangle in the world:

$$a^2 + b^2 = c^2$$

a^2	Area of the square on Side A
b^2	Area of the square on Side B
c^2	Area of the square on the Hypotenuse (the biggest side)

■ Section 5: Real-Life Example — The Ladder Problem

Imagine a ladder leaning against a wall. The **ladder** is the **hypotenuse** (the longest side), the **wall** is one shorter side, and the **ground** is the other shorter side. Together they form a perfect right-angled triangle!

■ If a ladder is 5 m long and its base is 3 m from the wall, how high up the wall does it reach? Use $a^2 + b^2 = c^2$!

Other real-life right-angled triangles you can spot:

- **A roof slope** — The slanted roof, wall, and ceiling form a triangle.
- **A ramp** — The ramp, ground, and vertical step form a triangle.
- **TV screen diagonal** — The width, height, and diagonal of a screen.
- **Map distances** — Finding the straight-line distance between two towns.

■ Engineers, architects, and builders use $a^2 + b^2 = c^2$ every single day!

■ Section 6: Step-by-Step Calculation

Let's solve the classic 3-4-5 triangle together. Riya and Arjun mark a right triangle on the classroom floor: one side is **3 m** and another is **4 m**. How long is the hypotenuse?

Step 1	Square each known side: $a^2 = 3^2 = 9$ $b^2 = 4^2 = 16$
Step 2	Add the two squares: $a^2 + b^2 = 9 + 16 = 25$
Step 3	$c^2 = 25$, so $c = \sqrt{25} = 5$
✓ Answer	The hypotenuse is 5 metres long!

■ 3, 4, 5 is called a **Pythagorean Triple** — three whole numbers that perfectly satisfy $a^2 + b^2 = c^2$!

⇒ ■ Section 7: Practice Problems

Now it's your turn! Use $a^2 + b^2 = c^2$ to answer these questions. Show your working step by step.

Q1. Find the hypotenuse when $a = 6$ and $b = 8$.

■ *Your working:*

■ Answer: $6^2 + 8^2 = 36 + 64 = 100 \rightarrow c = \sqrt{100} = 10$

Q2. Find the missing side b when $c = 10$ and $a = 6$.

■ *Your working:*

■ Answer: $b^2 = c^2 - a^2 = 100 - 36 = 64 \rightarrow b = \sqrt{64} = 8$

Q3. True or False: A triangle with sides 5, 12, 13 is right-angled?

■ *Your working:*

■ Answer: $5^2 + 12^2 = 25 + 144 = 169 = 13^2 \rightarrow \text{TRUE } \checkmark$

■ Section 8: Fun Quiz — Which is the Hypotenuse?

For each triangle below, identify the hypotenuse and verify that $a^2 + b^2 = c^2$ holds true.

Triangle	Sides	Hypotenuse	Check: $a^2+b^2=c^2$?
A	3, 4, 5	5	$9 + 16 = 25 \checkmark$
B	6, 8, 10	10	$36 + 64 = 100 \checkmark$
C	5, 12, 13	13	$25 + 144 = 169 \checkmark$
D	8, 15, 17	17	$64 + 225 = 289 \checkmark$

■ Hint: The hypotenuse is always the **largest** number and sits **opposite** the right angle!

■ What You Learned Today

- A right-angled triangle has one 90° corner.
- Its three sides are called Side A (a), Side B (b), and the Hypotenuse (c).
- The hypotenuse is always the longest side, opposite the right angle.
- The Pythagoras Theorem states: $a^2 + b^2 = c^2$
- We can use this formula to find missing side lengths in real life.
- Pythagorean Triples like 3-4-5, 6-8-10, 5-12-13 are whole-number solutions.

**Keep curious, keep exploring —
maths is full of wonderful surprises! ■**